Mortality Prediction via Age-Specific Band Selection

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Abstract

A novel mortality prediction framework, age-specific band selection, is proposed to borrow information from "neighboring" ages and train prediction models for each individual age in a mortality table. This idea is further extended to borrow information across multiple populations and three specific approaches are proposed, including a distance-based approach, an ensemble approach, and an ACF model-based approach. Extensive empirical studies with the Human Mortality Database are conducted to confirm their enhanced prediction accuracy compared with relevant benchmark models.

Keywords: Mortality Prediction; Human Mortality Database; Age-specific Band Selection; Clustering; K-Nearest Neighbor.

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1 Introduction

Actuaries face a significant challenge in obtaining accurate future mortality rate prediction for insurance product design, pricing, and many other risk management purposes. While the past decades have witnessed a proliferation of various mortality models, there is still a demand for improved prediction performance. This paper aims to contribute in this direction by establishing novel data-driven procedures that select pertinent mortality data to facilitate information borrowing and inform extrapolative modeling so as to improve the accuracy of forecasting.

Prominent among the available models is the Lee-Carter mortality projection model (Lee and Carter, 1992) and its extensions (Wilmoth, 1993; Lee and Miller, 2001; Booth et al., 2002; De Jong and Tickle, 2006). For senior age groups (e.g., 65 or larger), the Cairns-Blake-Dowd (CBD) model (Cairns et al., 2006) and its generalizations (e.g., Cairns et al., 2009) are widely recognized for their superior predictive performance. These models can be considered as vehicles for borrowing information across ages, and their calibration involves the specification of age ranges for mortality data. Recent literature underscores the impact of different age range choices on predictive results, such as the contrast between full-age range and partial-age range models conducted by Shang and Haberman (2020).

Analyzing mortality data from the Human Mortality Database reveals similarities in development patterns within certain age groups (e.g., young ages, middle ages, and senior ages) and some distinctions across these age groups. Building on the assumption that similarities of age-specific mortality patterns are conducive to improving mortality predicting accuracy, Tsai and Cheng (2021) incorporated statistical clustering methods into mortality models to improve predicting performance. Furthermore, Wang et al. (2021) considered the "neighboring" effect under the assumption that future mortality development of a target age is influenced by the development patterns of its "neighboring" ages.

This paper introduces a novel age-specific band (ASB) based mortality prediction framework. It dissects the overall prediction objective into separate predicting tasks for individual ages, and in each predicting task, we focus on the prediction of one individual age that we call the target age. Based on the assumption that mortality development patterns are similar in adjacent ages, it is desirable for a target age to borrow information from these neighboring ages to achieve prediction accuracy enhancement. The ASB framework considers borrowing information from age bands centered at the target age of different spans, and the chosen age band is expected to embrace, to the largest extent, the potential benefits of the similarity in mortality development patterns shared by the ages in the band and the target age. A smoothing procedure is further incorporated into the ASB framework to mitigate the risk of a substantial disparity in the band size between two consecutive target ages.

The introduced ASB-based mortality prediction framework is adaptable for integrating various existing mortality forecasting models. As an illustrative example, we utilize the Lee-Carter model as the embedded model in this study. To assess the performance of the proposed ASB-based prediction method, we analyze mortality data from 24 populations of both genders in the Human Mortality Database (HMD), alongside benchmarks utilizing statistical learning methods such as clustering. Empirical findings substantiate the effectiveness of the proposed ASB selection based approach, as it consistently enhances predictive accuracy across a majority of the 48 populations studied.

Leveraging information across populations has gained prominence and proven advantageous with appropriate methods, as demonstrated by prior studies (Li and Lee, 2005; Li and Hardy, 2011; Russolillo et al., 2011; Li, 2013; Zhou et al., 2014; Kleinow, 2015; Hyndman et al., 2013; Shang and Hyndman, 2017; Diao et al., 2021), among others. In this paper, we extend the ASB selection method to borrow information across multiple populations and introduce three specific approaches for implementation: a distance-based method, an ensemble method averaging outcomes from random subsets, and an ACF model-based approach fusing the age band framework with geographic data. Extensive empirical comparisons using the Human Mortality Database confirm improved predictive accuracy for all the three methods. Notably, the distance-based approach yields substantial accuracy enhancements, while the ensemble method matches its accuracy with significantly lower computational requirements. The ensemble method also exhibits robustness across various hyperparameter values. Additionally, empirical findings suggest that an ASB selection based on the ACF model performs better for female populations, while one based on the simpler Lee-Carter model is preferable for male populations.

The remainder of this paper is organized as follows. In Section 2, we present a comprehensive description of the proposed ASB based prediction framework, along with the data-mining procedures employed to determine the optimal age band selection. Section 3 showcases empirical studies of the proposed methodologies, comparing them against benchmark models, based on analysis with the Human Mortality Database. In Section 4, we introduce the three specific approaches designed to borrow information in a multi-population context. Subsequently, Section 5 delves into the results and findings obtained for multi-population data. Lastly, we conclude our paper in Section 6.

2 Age-Specific Band Method for Single-Population Data

In the prediction of a mortality table, the conventional approaches entail calibrating a mortality model using data from the entire age range and subsequently employing the calibrated model to generate predictions for all ages in the mortality table. However, this approach does not account for the potential dissimilarities in mortality development patterns that might exist among age groups. In response to this limitation, we propose the Age-Specific Band (ASB) based framework for mortality prediction in this section. Within this framework, we deconstruct the overall predictive objective into separate prediction tasks for individual ages in the mortality table. For each specific age, we design dedicated machine learning algorithms to identify a desirable age band, from which relevant information is drawn to enhance the mortality prediction accuracy specifically for the target age.

2.1 General Age-Specific Set Selection

Before we delve into the ASB based framework, we first describe a broader framework that selects a set of ages, not necessarily an age band, for the mortality prediction of each individual age. This concept is essential for us eventually to extend the ASB approach from the context of single-population to the situation with multiple-population data. Let $\mathcal{X} := \{x_L, x_L + 1, \ldots, x_U\}$ represent the age range in the mortality table that we aim to predict. Within this framework, we conduct mortality prediction for each individual age $x_0 \in \mathcal{X}$ separately. For each individual target age x_0, we select an age-specific set $\mathcal{A}^*_{x_0}$, which must contain the target age x_0, to be used in the calibration of a mortality model through a certain procedure that we will provide details later. Assuming that mortality data in the study lies in an age-year window $\mathcal{X} \times \mathcal{T}$ with $\mathcal{T} = \{t_L, t_L + 1, \cdots, t_U\}$. For target age $x_0 \in \mathcal{X}$, a mortality model is calibrated only using the data over the age-year window $\mathcal{A}^*_{x_0} \times \mathcal{T}$. The proposed framework is versatile and can be embedded with any single-population mortality forecasting models. When the Lee-Carter model or the CBD model is embedded, we do the calibration regarding the following formulation and data range:

$$\log m(x,t) = a_x + b_x k_t + \epsilon_{x,t}, \quad x \in \mathcal{A}^*_{x_0}, \ t \in \mathcal{T}, \tag{1}$$

or

$$\operatorname{logit} q(x,t) = \log\left[\frac{q(x,t)}{1-q(x,t)}\right] = K_t + (x - \bar{x})k_t + \epsilon_{x,t}, \quad x \in \mathcal{A}^*_{x_0}, \ t \in \mathcal{T}$$

where $m(x,t)$ denotes the central death rate, also known as the age-specific death rate, for age $x \in \mathcal{X}$ and year $t \in \mathcal{T}$ of a population, $q(x,t)$ is the probability that an individual

of a population aged $x \in \mathcal{X}$ will die between t and $t+1$ given the individual is alive at time $t \in \mathcal{T}$, and $\epsilon_{x,t}$'s are independent and identically residuals with zero mean and finite age-specific variance σ_x^2. The mortality prediction for the target age x_0 is then obtained by extrapolating the established mortality model. The prediction of the mortality table is formed by repeating the procedures of age set selection, model calibration, and extrapolation for every individual age. In the rest of the paper, we will focus on the Lee-Carter model as the embedded model in the proposed age-specific set selection framework due to its predominance in the literature for mortality prediction.

2.2 Age-Specific Band Method

One specific approach to define $\mathcal{A}_{x_0}^*$ for the target age x_0 is to utilize information from the neighboring ages of x_0. The rationale behind this approach is rooted in the observation that adjacent ages tend to exhibit more similarities in mortality development patterns compared to ages that are distant from the target age x_0. This observation prompts us to consider $\mathcal{A}_{x_0}^*$ as an age band encompassing the target age x_0 and its adjacent ages.

2.2.1 Symmetric Age Band

One convenient choice of the age band is a symmetric one centered at the target age x_0 defined below:

Definition 2.2.1 (**Symmetric Age Band**). *For a given target age x_0, a symmetric age band of x_0 with a band radius r, denoted as $\mathcal{A}_{x_0}(r)$, consists of ages from $\tilde{x}_L$ to $\tilde{x}_U$, where This age band spans from $\tilde{x}_L$ to $\tilde{x}_U$, where $\tilde{x}_L \leq x_0 \leq \tilde{x}_U$.*

$$\begin{aligned} \tilde{x}_L &= \max(x_0 - r, x_L), \\ \tilde{x}_U &= \min(x_0 + r, x_U). \end{aligned}$$

Note that the bound ages $\tilde{x}_L$ and $\tilde{x}_U$ in the above definition should be x_0 specific. We suppress these notations for simplicity of presentation thereafter unless otherwise stated.

As the name suggested, a symmetric age band includes an equal number of ages from both sides of the target age unless the target age is nearby the smallest or the largest age in the mortality data range. For a given target age x_0, different band radii r result in distinct age bands $\mathcal{A}_{x_0}(r)$, representing different extents of borrowing information from the neighboring ages. Therefore, the task of determining a desirable symmetric age band for each target age x_0 can be accomplished by determining the age-specific value of the hyperparameter r, denoted as $r_{x_0}^*$. It is well-known that the mortality development patterns

between two adjacent ages are highly resemblant. Therefore, we do not expect that the value of $r^*_{x_0}$ varies much when x_0 slides within a small neighborhood.

2.2.2 Selection of Age Band

As discussed previously, age band selection is essentially a procedure of determining the value of the hyperparameter r for symmetric age bands. In principle, any validation procedure can be adopted to determine an optimal parameter. To fully leverage the available mortality data, we use a block cross-validation method, inspired by Bergmeir and Benítez (2012) and Bergmeir et al. (2014). The validation method is an adaption of conventional cross-validation tailored for time series data. As pointed out by Bergmeir et al. (2014), this method maximizes the utilization of data, and in the meanwhile, ensures a certain degree of independence by maintaining a temporal separation between the training and validating blocks. The method has recently gained attention in the mortality prediction literature and was used as a robust tool for obtaining validation error estimates; see Kessy et al. (2021) and SriDaran et al. (2022).

To conduct the block cross-validation procedure, the available training data is partitioned into u blocks of roughly equal size sequentially. Each block of sequential data is used as the validation set once to evaluate the performance of trained models, while the remaining $u-1$ blocks are used as the training blocks, with s-year data are omitted from both sides to the validating block. The omission enhances some degree of independence between the training and the validating data. The selection of u is a trade-off between the computational cost and data availability, where a larger u entails more model estimations but a smaller training set. Similarly, the choice of s, that measures the size of omitted data, balances the level of independence and the data availability for training. A large s may lead to a considerable loss of data and even potentially result in insufficiency of data for model estimation, while a small s could compromise the approximate independence between the training and the validating data. Figure 1 visualizes how data is partitioned into training and validating blocks for a 4-fold block cross-validation in the empirical study in Section 3.

For each validating block, we calibrate the Lee-Carter model (1) using a standard SVD procedure on the corresponding training block to obtain the fitted values of age-specific coefficients a_x and b_x and the period index k_t, and fit a random walk with drift

$$k_t = k_{t-1} + d + \epsilon_t,$$

where d serves as the drift parameter and ϵ_t as the noise term. Either a forward fill or backward fill, shown in equations (2) and (3), respectively, is then utilized to project k_t

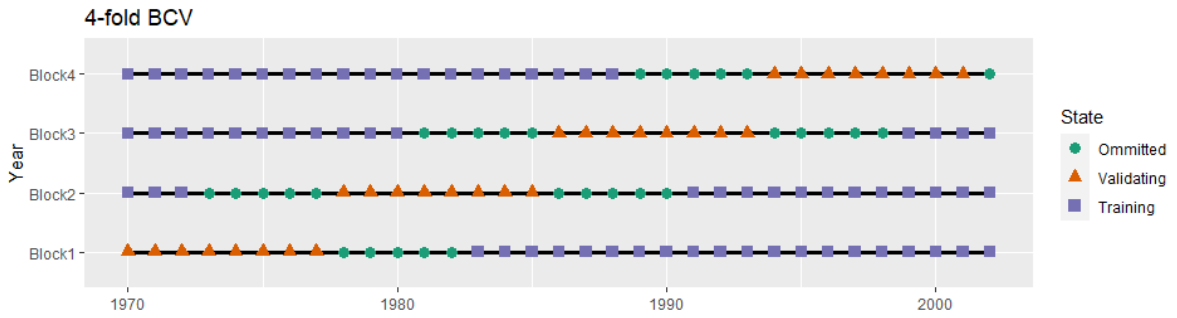


Figure 1: Illustrative figure of a 4-fold block cross-validation with $s = 5$. Purple squares represent points from the time series used for model training, orange triangles represent points used for validation, and green circles represent the omitted data points.

into $\mathcal{T}_{validate}$ based on estimated $\hat{d}$ and the "closest" realized values in $\mathcal{T}_{train}$:

$$\textbf{forward fill}: \hat{k}_t = k_{t-\alpha} + \alpha\hat{d}, \tag{2}$$

$$\textbf{backward fill}: \hat{k}_t = k_{t+\alpha} - \alpha\hat{d}. \tag{3}$$

In the above equations, $\hat{k}_t$ represents the projected value at time $t \in \mathcal{T}_{validate}$ and $k_{t-\alpha}$ or $k_{t+\alpha}$ represents the "closest " realized values of the sequence in the training period. The forward fill approach is often used so that the values of k_t in $\mathcal{T}_{validate}$ are predicted based on $k_{t-\alpha}$, the latest realized value in the preceding training period using equation (2). However, if the validation block is the first block in the temporal order, the values of k_t in $\mathcal{T}_{validate}$ are projected from $k_{t+\alpha}$, the first realized value in the successive training period, using equation (3).

Let $\mathcal{T}^{(v)}_{validate}$ denote the set of years spanned by the v-th validation set, and $\mathcal{T}^{(v)}_{training}$ be the set encompassing all the years involved by the corresponding training data. Then, the block cross-validation sum of squared error (BCV-SSE) for a given age band $\mathcal{A}_{x_0}(r)$ is calculated as

$$\text{BCV-SSE}(\mathcal{A}_{x_0}(r)) = \sum_{v=1}^{u} \sum_{t \in \mathcal{T}^{(v)}_{validate}} \left[\log m_{x_0,t} - \log \hat{m}^{(v,r)}_{x_0,t}\right]^2, \tag{4}$$

where $m_{x_0,t}$ denotes the central death rate for age x_0 and year t, and $\hat{m}^{(v,r)}_{x_0,t}$ is the respective predicted value from the mortality model trained using data from $\mathcal{A}_{x_0}(r) \times \mathcal{T}^{(v)}_{training}$.

We choose the best age band $\mathcal{A}^*_{x_0}(r^*_{x_0})$ with the band radius $r^*_{x_0}$ achieving the smallest BCV-SSE, i.e.,

$$r^*_{x_0} = \text{argmin}_r \text{BCV-SSE}(\mathcal{A}_{x_0}(r)) \ .$$

The above model fitting and cross-validation procedure are described for the Lee-Carter

model and can be similarly adapted to other mortality models, e.g., the CBD model.

2.2.3 Smoothing

Implementing the block cross-validation procedure for different target ages generates a sequence of optimal radii $r^*_{x_0}$'s. The inherent randomness of the realized BCV-SSE values inevitably introduces variability in the results, and it could lead to roughness and bumpiness in the $r^*_{x_0}$ sequence as the target age x_0 varies. One should expect that the chosen age bands of adjoining target ages are not largely distinct and incomparable, and thus it is reasonable to require the $r^*_{x_0}$ sequence to have some degree of smoothness. We apply a smoothing procedure to the $r^*_{x_0}$ sequence via a "moving average" scheme, where we replace the radius with an average of some adjacent points in the original sequence:

$$r^*_{x_0.\mathbf{smooth}} = \frac{1}{2h+1}\left(r^*_{x_0-h} + r^*_{x_0-h+1} + \cdots + r^*_{x_0} + \cdots + r^*_{x_0+h}\right)$$

where h is known as the radius of the smoothing window. For target ages close to the boundaries x_L or x_U, the averaging is applied to a truncated section of the array, from $r^*_{x_L}$ to $r^*_{x_0+h}$, or from $r^*_{x_0-h}$ to $r^*_{x_U}$. In the empirical study of Section 3, we will show the effect of different choices of h.

3 Empirical Study with Age-Specific Band Method

In this section, we assess the performance of the proposed ASB based prediction method empirically by analyzing data from the Human Mortality Database (HMD) and conducting comparisons with the conventional Lee-Carter model and the clustering-based methods as proposed in Tsai and Cheng (2021).

3.1 Empirical Setting and Predictive Models

We consider the mortality data from 24 populations as listed in Table 1 for $\mathcal{X}\times\mathcal{T}$, where $\mathcal{X} = \{0, 1, \cdots, 100\}$ and $\mathcal{T} = \{1970, 1971, \cdots, 2010\}$. The geographic information presented in the table will not be used in this section but will be used in Section 5. In our study, this dataset is divided into a training set, spanning 1970 to 2002, and a test set encompassing 2003 to 2010. For each prediction model, we utilize the training dataset (1970–2002) to train the model and subsequently extrapolate the trained model into the testing period (2003–2010) to obtain mortality prediction.

The prediction models used for comparison in this empirical study include:

Table 1: 24 populations from HMD and their geographic information.

Target Population	Geographic Group	Target Population	Geographic Group
Australia	Oceania	Netherlands	West Europe
Austria	West Europe	New Zealand	Oceania
Japan	Asia	Norway	Scandinavia
Belgium	West Europe	Poland	East Europe
Scotland	Great Britain	Portugal	South Europe
Canada	North America	U.S.A.	North America
Czech Republic	East Europe	Slovakia	East Europe
Denmark	Scandinavia	Spain	South Europe
Finland	Scandinavia	Sweden	Scandinavia
France	West Europe	Switzerland	West Europe
Hungary	East Europe	Taiwan	Asia
Italy	South Europe	England & Wales	Great Britain

- **Lee-Carter**: As a benchmark model, the Lee-Carter model is fitted to each population and gender separately with the full-age band from 0 to 100.
- **ASB Based on Lee-Carter Model**: The ASB based approach embedded with the Lee-Carter model is applied for the mortality prediction of each population.

  To adopt the block cross-validation procedure, the training data, spanning 1970 to 2002 for a length of 33 years, is partitioned into four non-overlapping blocks of 8 years plus the last year of 2002; see Figure 1 for a graphical illustration. Each of the four 8-year blocks, represented by the orange triangles in Figure 1, serves as a validating block in turn to evaluate the performance of prediction models with different age bands. Data within the closest $s = 5$ years to both sides of the validating block, shown as green circles in Figure 1, are omitted. The remaining data after omission, shown as purple squares in Figure 1, are then used to train the prediction model, and the final choice of optimal age band for a target age is determined based on the collective performance across all validating blocks.

  Based on whether a smoothing step is incorporated in the process of determining band radius, there are two distinct ASB based prediction models:

  - **LC-ageband**: Use the aforementioned 4-fold block cross-validation procedure to search for the optimal age band among all symmetric candidate age bands featured with different band radii, including both the age band containing the target age x_0 itself only and the full-age band including all ages from 0 to 100.

Repeat this for each age.

- **LC-ageband-smooth**: A smoothing step as described in Section 2.2.3 with the length of smoothing window $h = 4$ is applied upon the age band radii determined in the **LC-ageband** model.

- **Clustering-based models**: The clustering-based methods conduct a clustering analysis to group ages and apply the Lee-Carter model to each resulting cluster of ages (Tsai and Cheng, 2021). Specifically, we let $Y_{x,t}$ be the annual increment in logarithimic mortality rate of age x from year $t-1$ to year t, i.e.,

$$Y_{x,t} = \log m(x,t) - \log m(x,t-1), \quad t \in \{t_{L+1}, t_{L+2}, \ldots, t_U\}. \tag{5}$$

For a given age x, the sequence $Y_{x,t}, t \in \{t_{L+1}, t_{L+2}, \ldots, t_U\}$ is assumed as an independent sample from a normal distribution, and the location and variability of the sequence is characterized by mean μ_x and variance σ_x^2, respectively. When the pairs $(\mu_{x_1}, \sigma_{x_1}^2)$ and $(\mu_{x_2}, \sigma_{x_2}^2)$ display similarity, the two ages x_1 and x_2 are deemed suitable to be put within the same cluster. In light of this, the clustering algorithms are executed on a set of two-dimensional objects $\{(\hat{\mu}_x, \hat{\sigma}_x^2) : x \in \mathcal{X}\}$, where the estimated values are computed as:

$$\hat{\mu}_x = \frac{1}{U-L} \sum_{i=L+1}^{U} Y_{x,t_i}, \tag{6}$$

$$\hat{\sigma}_x^2 = \frac{1}{U-L-1} \sum_{i=L+1}^{U} \left(Y_{x,t_i} - \hat{\mu}_x\right)^2. \tag{7}$$

Depending on the particular clustering method employed, three distinct prediction models are applied. Their model labels and the associated clustering methods are explained as follows:

- **LC-Kmeans**: The K-means clustering method is employed to establish age clusters. We use the recommended number of clusters from `NbClust` function in `R` package `NbClust` (Charrad et al., 2014).
- **LC-HCluster**: The Ward's agglomerative clustering method (Ward Jr, 1963) is applied to establish age clusters. This method employs a bottom-up merging approach using the sum of squared errors as the objective function on the two-dimensional objects $(\hat{\mu}_x, \hat{\sigma}_x^2)$. The optimal number of clusters is determined by the recommendation from the `NbClust` function in the `R` environment (Charrad et al., 2014).
- **LC-GMM**: The Gaussian mixture model (GMM) clustering method is utilized

to determine age clusters. The GMM approach assumes a mixture of Gaussian distributions for the data. Through iterative adjustments of the parameters of the mixture distribution to best align with the underlying data, each component distribution of the mixture models represents a distinct cluster. The GMM clustering algorithm comprises three key elements: (1) Initialization through clustering results from agglomerative hierarchical clustering method; (2) Maximum likelihood estimation using the EM algorithm; (3) Model selection and determination of the number of clusters via approximate Bayes factors with the BIC. We adhere to the strategy introduced by Fraley and Raftery (2002) for fitting data with a Gaussian mixture structure, employing the `mclust` function within the `R` package `mclust` (Scrucca et al., 2016). Further details about this method can be found in Fraley and Raftery (2002) and Tsai and Cheng (2021).

3.2 Predicting Performance

For a given population, we let $e(t)$ denote its age-aggregated test SSE and caculate it by

$$e(t) = \sum_{x \in \mathcal{X}} \left[\log m(x,t) - \log \widehat{m}(x,t)\right]^2, \quad t = 2003, \ldots, 2010, \tag{8}$$

where $m(x,t)$ represents the actual mortality rate and $\widehat{m}(x,t)$ denotes its predicted counterpart. Here $\mathcal{X} = \{0, 1, \ldots, 100\}$ is the full range of ages. The overall prediction accuracy is assessed through overall test SSE, which is computed as $\sum_{t=2003}^{2010} e(t)$. This measures the mortality prediction accuracy for one population. For a comprehensive comparison of prediction accuracy among different models, we compute the overall test SSEs for all 24 populations and report the 1st quartile, median, sample mean, and 3rd quartile of the resulting SSEs in Table 2. Smaller values of these statistics in this context imply better prediction performance. The nubmers in Table 2 reveal that the proposed ASB based method with smoothing (LC-ageband-smooth) significantly outperforms all other methods, evidenced by its smallest 1st quantile, median, sample mean and 3rd quantile of the resulting test SSEs for both female and male populations. The ASB based method without smoothing (LC-ageband) exhibits comparable predictive power with the LC-GMM method, which is the best among the three considered clustering-based methods — a result that aligns with empirical studies conducted with mortality data in the US and the UK for both genders by Tsai and Cheng (2021).

Additionally, we utilize one-sided Diebold-Mariano (DM) test, as introduced in Diebold and Mariano (1995) and Harvey et al. (1997), to compare the predictive accuracy of pairs of models. One-sided Diebold-Mariano (DM) test assesses the null hypothesis that one model is no better than the other and a p-value of 0.05 or less suggests that the null hypothesis is

Table 2: Summary statistics of test SSEs of 24 female populations and 24 male populations comparing the prediction performance of different predicting methods.

	1st Quartile	Median	Mean	3rd Quartile
Female Population				
Lee-Carter	19.37	44.73	44.64	63.90
LC-GMM	16.80	44.44	45.18	66.18
LC-Kmeans	18.01	46.84	45.15	71.06
LC-HCluster	16.99	45.43	45.02	72.40
LC-ageband	17.44	44.49	44.00	67.36
LC-ageband-smooth	14.81	41.94	40.57	59.22
Male Population				
Lee-Carter	28.40	41.31	49.79	67.97
LC-GMM	24.70	35.98	40.60	59.77
LC-Kmeans	27.21	38.55	45.97	62.67
LC-HCluster	27.15	38.92	46.30	62.67
LC-ageband	16.78	37.67	39.14	60.66
LC-ageband-smooth	16.01	34.99	37.44	53.90

rejected and one model is significantly superior to the other. We divide the six predictive models described in Section 3.1 into two groups: group one comprised of the two ASB based methods and group two consisting of all benchmark methods, i.e., the Lee-Carter model and the three clustering-based models. We compare the prediction accuracy of a pair of models, one from each group, by conducting two one-sided DM tests based on age-aggregated test SSEs from two comparing models. One test assesses the null hypothesis that the model from the first group is no better than the one from the second group and the other test evaluates the null hypothesis that the model from the second group is no better than the one from the first group. If the former null hypothesis is rejected, we conclude that the model from group one is better than the one from group two and thus the model from group one wins the comparison; if the latter null hypothesis is rejected, the model from group two wins; if none of the two hypotheses are rejected, no model wins the comparison. For each pair of models, winner determination is repeated for all 24 populations and a higher frequency of winning indicates a superior prediction performance. In Table 3, we report the comparison results between two groups of models, the ASB based models listed in the row and the benchmark models listed in the column. In each cell of Table 3, we report two integers, of which the first is the number of wins by the model in the row, and the second is the number of wins by the model in the column. out of 24 populations. Table 3 shows that the ASB based models notably outperform the benchmark models in delivering more accurate predictions evidenced by their dominating numbers of wins. This advantage is particularly evident for the ASB based model with a smoothing procedure.

Table 3: Number of wins for comparisons between the proposed models (in the row) and the benchmark models (in the column) based on pairs of one-sided DM tests. In each cell, the first integer indicates the number of wins of the model in the row and the second integer is the number of wins of the model in the column out of 24 comparisons.

	Lee-Carter	LC-GMM	LC-Kmeans	LC-HCluster
Female Population				
LC-ageband	$(10, 4)$	$(10, 6)$	$(13, 5)$	$(12, 5)$
LC-ageband-smooth	$(17, 0)$	$(19, 1)$	$(19, 1)$	$(17, 1)$
Male Population				
LC-ageband	$(18, 1)$	$(11, 4)$	$(15, 1)$	$(15, 2)$
LC-ageband-smooth	$(22, 0)$	$(19, 3)$	$(19, 0)$	$(17, 0)$

We are also interested in examining how the performance of the proposed ASB based methods varies across age groups. To the end, we divide the entire age spectrum into six non-overlapping age groups as follows:

- **Child:** ages 0 to 9;
- **Teenage:** ages 10 to 19;
- **Young Adult:** ages 20 to 29;
- **Adult:** ages 30 to 59;
- **Retiree:** ages 60 to 89;
- **Elder:** ages larger than 89.

We compute the age-aggregated test SSEs defined in (8) over a refined age group, i.e., with $\mathcal{X}$ in the formula (8) taken as one of the above age groups, and calculate the overall test SSE accordingly. We depict boxplots of overall test SSEs (logarithmically scaled) of 24 populations in Figures 2 and 3. Subfigures in the left panels correspond to female populations and the ones on the right correspond to male populations. Subfigures in Figure 2 provide results for Child, Teenage, and Young Adult age groups and those in Figure 3 for Adult, Retiree and Elder age groups. In these graphs, a boxplot leaning toward a leftward position indicates greater prediction accuracy. As depicted in Figures 2 and 3, the degree of improvement in predicting accuracy by using the proposed ASB based methods varies across genders and age groups. For female populations, significant improvement in predicting accuracy is observed for the "Young Adult", "Adult", and "Retiree" age groups, which encompass ages 30 to 89. However, the predicting performance is comparable with other methods for other age groups. For male populations, all ages, except for the "Child" (ages 0 to 9) and "Elder" (ages above 89) groups, exhibit a noticeable reduction in test SSEs due to the adoption of the proposed ASB-based methods. There is a slight deterioration in prediction performance for the ASB based method for the "Elder" age group.

Notably, the scale of the horizontal axis varies across different age groups. Specifically, it is significantly smaller for the "Elder" group (ages above 90) compared to the other groups. Therefore, a visually apparent difference in the positioning of the boxplots between two models for the "Elder" group does not necessarily imply a substantial distinction in their test SSEs. This scale variation should be kept in mind while interpreting the relative performances of prediction models for different age groups upon Figures 2 and 3.

3.3 Influence of Smoothing

The results in Section 3.2 have demonstrated an encouraging improvement in the overall prediction accuracy achieved through the smoothing step when the smoothing window is

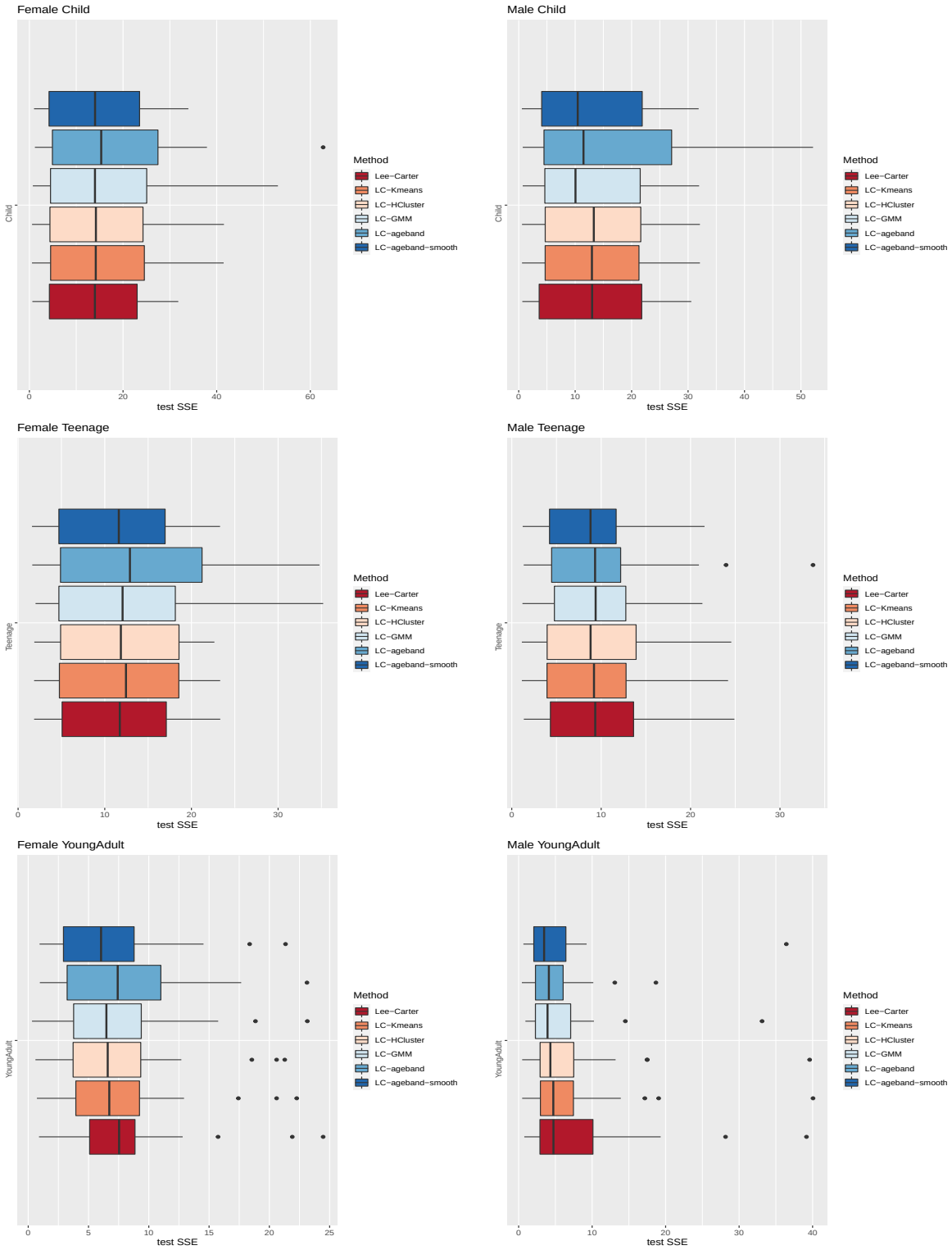


Figure 2: Boxplots of overall test SSEs of 24 female populations (left panel) and 24 male populations (right panel) comparing the prediction performance of different methods for different age groups: Child, Teenage, and Young Adult.

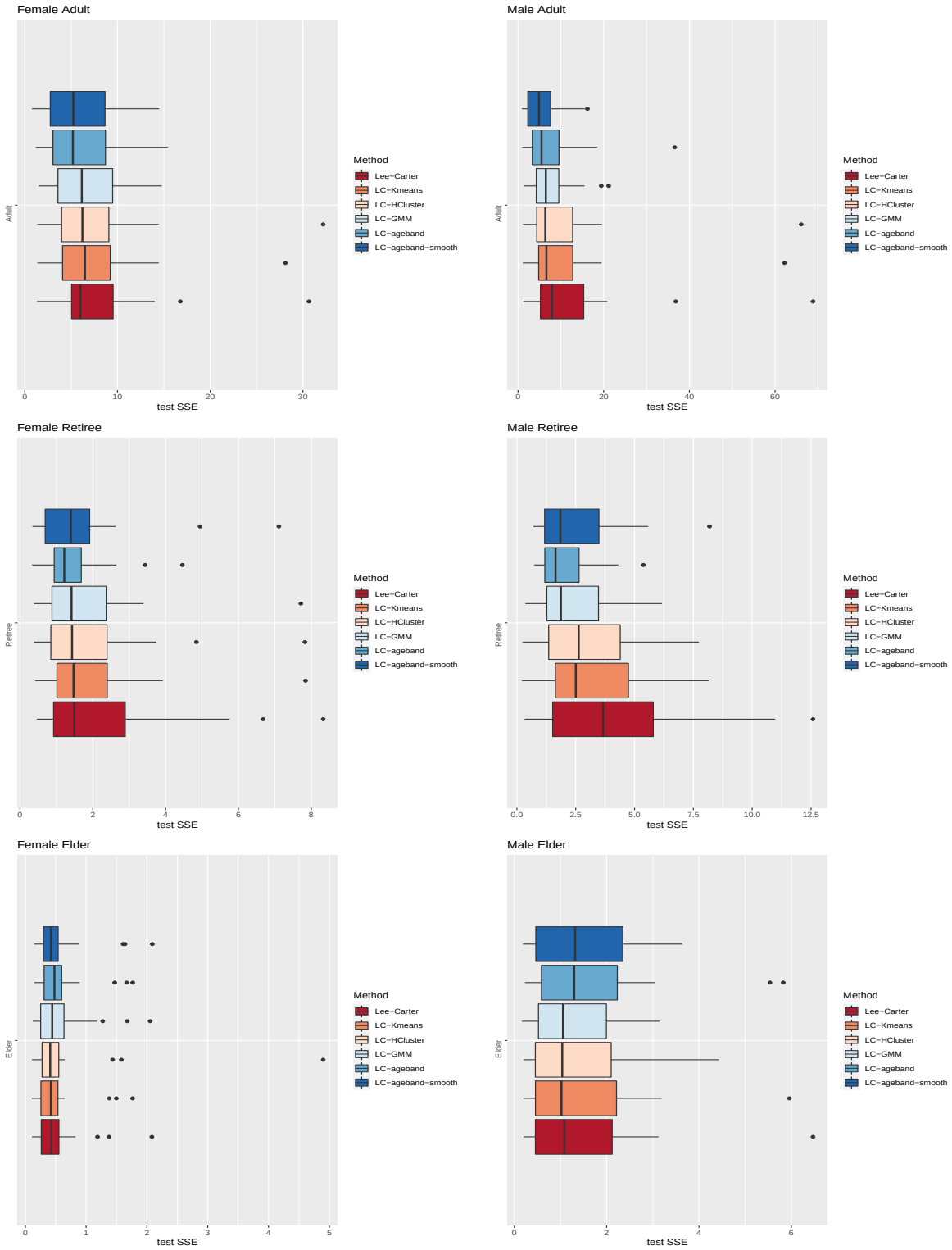


Figure 3: Boxplots of overall test SSEs of 24 female populations (left panel) and 24 male populations (right panel) comparing the prediction performance of different methods for different age groups: Adult, Retiree, and Elder.

set to $h = 4$. In this section, we study how varying degrees of smoothness, corresponding to different values of h, influence the extent of the improvement in prediction performance.

Figure 4 shows how averaged test SSEs over 24 populations change with respect to different values of smoothing window h for both genders. A wide range of smoothing window lengths, representing scenarios of no smoothness ($h = 0$), weak smoothness ($0 < h < 3$), moderate smoothness ($3 \leq h < 10$), and strong smoothness ($h = 20$ or even $h = 50$), are considered. We also include the results of the Lee-Carter model and another smoothing scheme based on the median, known as "3RS3R" (Tukey et al., 1977), for comparison. Figure 4 suggests that the average test SSEs fluctuate lightly and form a flat bottom over the range of moderate smoothness ($3 \leq h < 10$), and these most favourable average test SSEs are dramatically smaller than the one of the Lee-Carter model and further improve upon the median smoothing technique "3RS3R". We thus recommend incorporating smoothing into the proposed ABS method with moderate smoothness.

Another question of interest is whether there is a discernible pattern in the selected band radius as the target age varies from 0 to 100. To address this question, we calculate the average of the smoothed optimal band radius $r^*_{x_0.\mathbf{smooth}}$ over 24 populations and illustrate the values with respect to the target age in Figure 5. Figure 5 clearly demonstrates that smaller band radii are preferred for middle ages, whereas larger band radii are favoured for younger and elder ages. This suggests that middle ages can benefit from borrowing information from a moderate number of neighbouring ages to improve their mortality prediction accuracy, while younger and elder ages typically require more information and thus tend to borrow from a broader range of ages.

4 Age-Specific Set Methods for Multi-Population Data

In this section, we aim to make prediction for multiple populations based on the combined information from multiple mortality tables. We extend the foundational concept of the ASB based method introduced in Section 2 to multiple-population data, and similarly chunk the overall goal into separate prediction tasks for each individual age. We introduce three methods tailored for this purpose, each aiming to leverage information across ages and populations to refine mortality forecasts for an individual age.

4.1 Distance-based Approach

The overall prediction objective is broken down into micro goals of predicting for an individual age of a given population based on data for multiple populations. For a target age of

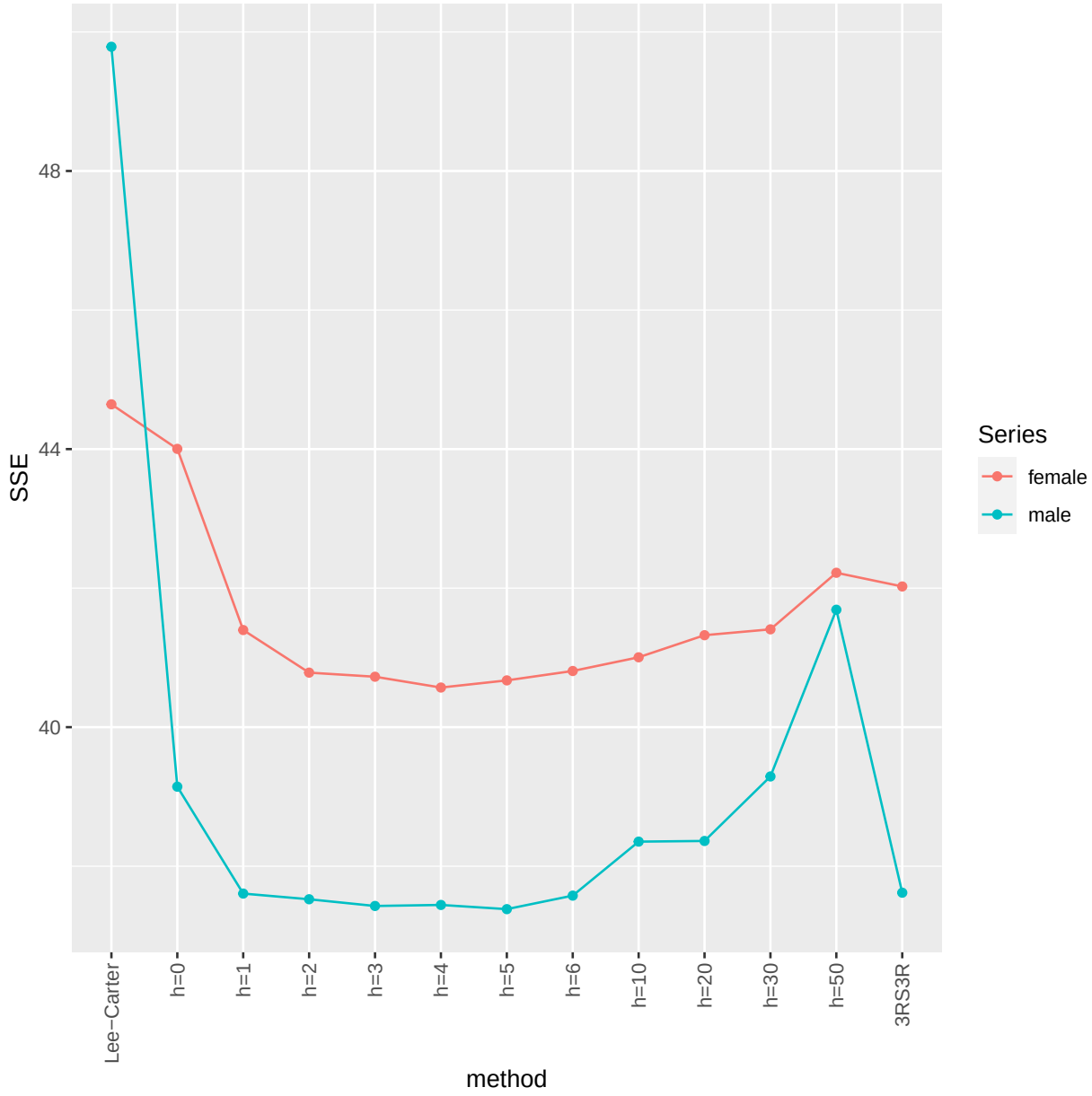


Figure 4: The Averaged values of test SSEs of 24 populations versus smoothing windows h.

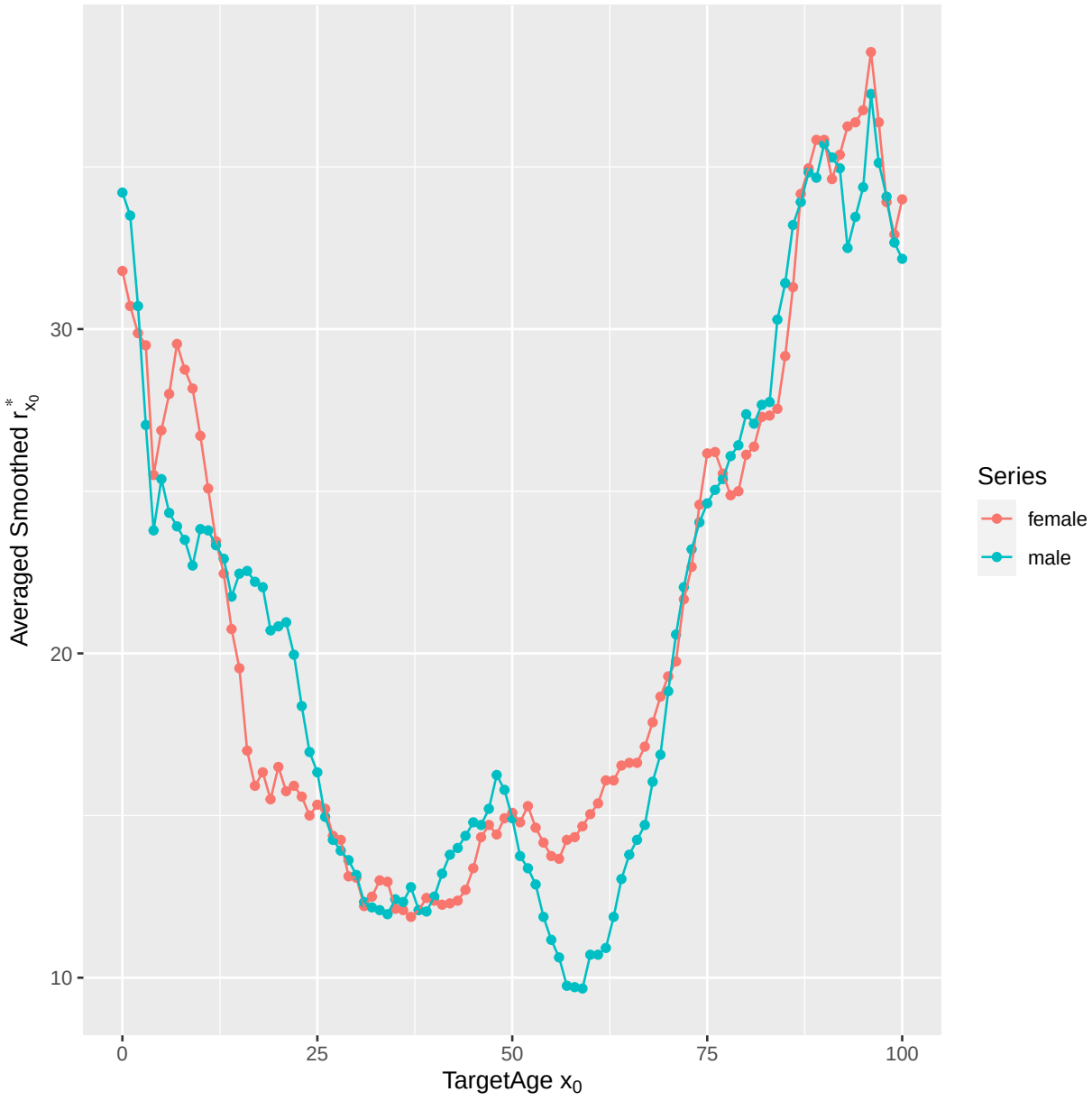


Figure 5: The averaged values of smoothed optimal band radius $r^*_{x_0.\mathbf{smooth}}$ versus target ages.

a target population, we aim to borrow information from a large information pool containing the mortality development trajectories of all ages other than the target age in the target population and those of all ages in other populations. The devised method must be adept at uncovering useful information and screening out noises to best assist the prediction for the target age, and in the meanwhile, maintaining computational efficiency. To achieve this, we extend the age-specific set paradigm introduced in Section 2.1 to multiple populations. In a single-population setting, we simplify the age-specific set method to the age-specific band method as discussed in Section 2.2 utilizing information in neighbouring ages. However, unlike the single-population scenario where the notion of "neighborhood" holds an intuitive interpretation due to the natural age ordering, defining a "neighborhood" for a target age in the entire information pool necessitates careful consideration. In this section, we propose a variance-based metric to quantify distance and delineate the composition of the "neighborhood", thereby expanding the age-specific set paradigm to information borrowing from multiple populations.

We first propose a distance measure to capture similarities in mortality developing trends. Two parallel mortality trajectories at different mortality levels are considered to have the same mortality developing trend and, therefore, the proposed measure is designed to ignore the differences led by mortality levels and focus on the movement over time. A general distance measure between two mortality trajectories is designed via the following steps.

1. Define a difference sequence as

$$\text{diff}_{x,i;y,j}(t) = \eta_{x,i}(t) - \eta_{y,j}(t), \quad t \in \mathcal{T},$$

   where $\eta_{x,i}(t)$, $t \in \mathcal{T}$ represents a mortality trajectory for age x in the ith population and the choice of $\eta_{x,i}$ depends on the chosen mortality model. $\eta_{x,i}(t)$ can take the form of $\log m(x,i,t)$ for Lee-Carter model and $\text{logit}\, q(x,i,t)$ for CBD model. "$\text{diff}_{x,i;y,j}(t)$" represents the difference between the mortality sequence of age x in population i and that of age y in population j.

2. Define a dissimilarity measure $D_{x,i;y,j}$ by the variance of the difference sequence, i.e.,

$$D_{x,i;y,j} = \text{Var}\left(\text{diff}_{x,i;y,j}(t)\right).$$

A lower value indicates a greater resemblance of the two mortality sequences and a closer alignment in their mortality development patterns over time. $D_{x,i;y,j} = 0$ for two parallel sequences which have a constant difference over time.

For a target age x_0 of a given population i_0, an age-and-population specific set $\mathcal{A}^*_{x_0,i_0}$ is

a set of ages potentially from multiple populations and the respective mortality sequences associated with the ages in $\mathcal{A}^*_{x_0,i_0}$ are used to calibrate mortality model and make prediction for age x_0 of population i_0. We consider $\mathcal{A}_{x_0,i_0}$ as a set of "neighboring" ages and the formation of the "neighborhood" is grounded in the way we define the distance. The distance measure $D_{x_0,i_0;y,j}$ for every y in the full age range and every population j is computed and the sequences are sorted in increasing order according to their distances to the target age of the target population. $\mathcal{A}^*_{x_0,i_0}(k)$ is a set of k nearest "neighbors", which echoes the well-known k-nearest neighbors algorithm (KNN)(Fix and Hodges (1989)). k represents the size of the "neighborhood" and reflects the extent of information borrowing. Let k take possible values in $\{0, 1, \cdots, K\}$, where K represents the maximum allowed number of sequences to borrow information from. Selecting the optimal age-and-population specific set reduces to selecting the optimal 'neighborhood" size. A 4-fold block cross-validation procedure, similar to the one used for the selection of the optimal radius r for the age-specific band method detailed in Section 2.2, is used to determine the optimal "neighborhood" size k, which we denote as $k^*_{x_0,i_0}$. Similar to the argument in Section 2.2.3, a smoothing step is advisable to mitigate the potentially abrupt fluctuations in the size of $\mathcal{A}^*_{x_0,i_0}$ as the target age x_0 varies over the full age range. In the empirical studies, we employ the same moving average smoothing scheme described in Section 2.2.3 with a smoothing window $h = 4$.

The predetermined value of the maximum permissible number of ages, K, can impact the performance of the proposed distance-based approach. A larger K value provides a greater potential for improving predictions by leveraging a broader candidate pool but introduce a heavier computational burden. Conversely, a smaller K value assumes that the most valuable information is confined within ages with strong similarities only. In the empirical studies discussed in Section 5, we explore the options of $K = 100$, 200, and 400 to study the marginal improvements in prediction accuracy achieved by increasing K values.

The description above pertains to the multi-population distance-based approach. If the approach is adapted to borrow information solely from ages within the target population, it is referred to as the single-population distance-based approach.

4.2 Ensemble Approach

As observed later in Section 5.1, an increase in the maximum allowed number of ages K to form a richer candidate pool for selection can enhance predictive performance. However, this gain comes at the cost of increased computational demand. To strike a balance between prediction improvement and computational feasibility, we propose an ensemble approach in this section. Inspired by Diao et al. (2023), the proposed ensemble approach consists of two main steps:

1. **Base leaners construction.** To establish a base learner, we randomly selected a subset of a moderate size n from the complete pool of sequences for all ages and populations, evaluate the distance from the sequences in the selected subset to the target sequence, sort the distances and form a "neighborhood" of a pre-specified size m ($m \leq n$), and calibrate a mortality model using the sequences in the "neighborhood". We repeat this random subsampling process N times and build N base learners, each generating a mortality prediction for the target age x_0 of the target population i_0.
2. **Result aggregation.** The N predictions can be aggregated using various averaging techniques to obtain a final prediction. For simplicity, we adopt a straightforward arithmetic average scheme.

In the implementation of the ensemble method, there are three key hyperparameters:

- N: The number of times the random subsampling is repeated.
- n: The size of each randomly chosen subset.
- m: The "neighborhood" size for information borrowing within each subset.

Different combinations of values for the three hyperparameters and their influence on predictive performances are studied in Section 5.3. Unlike the distance-based method, the ensemble approach does not require cross-validation, thus significantly reducing the computational time. The ensemble approach presents a pragmatic practice, balancing the need for enhancing prediction accuracy and alleviating the computational intensity.

4.3 ACF Model-based Approach

A factor worth considering for further improving the effectiveness of borrowing information across ages and populations is the choice of the underlying mortality model. In Sections 4.1 and 4.2, Lee-Carter model is embedded in the distance-based approach and ensemble approach, respectively. In this subsection, we consider an approach embedded with the ACF model (Li and Lee, 2005), a multiple-population mortality model.

The age set selection comes with various options. Instead of making a selection from a complete pool containing all ages and populations, we consider selecting the age set and population separately based on different criteria via separate procedures. As for population selection, we divide populations into geographic regions based on exogenous geographic information and use the ones in the same geographic region for predicting the mortality of a target population. The geographic division of populations has been a standard practice in multi-population mortality modeling literature; see, for instance, Richman and Wuthrich (2018). Populations in the same geographical region are considered to have more similar mortality trends and thus are more appropriately modelled jointly by a multiple-population

model, for instance, the ACF model. As for the age set selection, we adopt the age band selection paradigm introduced in Section 2. More specifically, we consider a sequence of age bands of different radii for the target age x_0, and employ the block cross-validation procedure to determine the optimal band radius. A smoothing step of the moving average scheme can be applied to obtain the smoothed age-specific band radii. After selecting the population and age band, an age-specific ACF model is built for every target age of the target population, and the prediction of mortality is then made based on a projetion using the resulting ACF model.

5 Empirical Studies with Multi-Population Mortality Data

In this section, we empirically evaluate the performance of the three proposed approaches in Section 4 using the same mortality data from the HMD analyzed in Sections 3. We apply the same partition of the data into a training set and a test set as in Sections 3. The present section consists of three subsections. Sections 5.1 compares the performance of the distance-based approach with some benchmark models. Section 5.3 compares results among the three proposed approaches. Section 5.2 investigates the impact of hyperparameters in the ensemble approach.

5.1 Performance Evaluation for the Distance-based Approach

As previously discussed, the distance-based approach can be employed to borrow information within a single target population or across multiple populations. In this empirical study, we consider the following two single-population distance-based models, which borrow information from a candidate pool of ages 0-100 within the target population.

- **LC-distance**: For a target age x_0, the other ages are ranked according to their distances to x_0 based on the proposed distance measure, instead of the natural numerical order of the ages. The optimal age set $\mathcal{A}^*_{x_0}$ contains $k^*_{x_0}$ nearest "neighboring" ages, where $k^*_{x_0}$ is selected by a 4-fold block cross-validation. The maximum allowed "neighborhood" size is set to $K = 100$.
- **LC-distance-smooth**: A moving averaging smoothing procedure with a smoothing window of length $h = 4$ is applied to the optimal "neighboring" sizes obtained from **LC-distance**.

We also explore two multi-population distance-based models, which borrow information across all 24 populations and thus form a pool containing ages 0-100 from each of these 24

populations, totalling $2424(= 101 \times 24)$ candidates.

- **Multi-LC-distance**: This method is implemented similarly to **LC-distance** except for ranking the distances of $2423(= 2424 - 1)$ candidate ages to the target age x_0.
- **Multi-LC-distance-smooth**: A smoothing procedure with $h = 4$ is applied to the optimal "neighboring" sizes obtained from **Multi-LC-distance**. This method is implemented with $K = 100$, 200 and 400 to study the increase in prediction accuracy with respect to an increased maximum information borrowing capacity.

The overall prediction accuracy is evaluated by the overall test SSEs as in Section 3. Table 4 reports summary statistics of test SSEs from the 24 populations for the distance-based approached mentioned in this section. We also include the results of the Lee-Carter model, LC-ageband and LC-ageband-smooth for comparison. We have the following interesting observations.

1. Single-population distance-based model and the age-specific band method yield comparable prediction results. This can be seen by comparing the rows "LC-distance" versus "LC-ageband" and "LC-distance-smooth" versus "LC-ageband-smooth" for both male and female populations. This result is expected since (1) both models make predictions based on information within the target population and (2) ages closer to the target age tend to have more similar development trajectories over years, consequently, the selected age sets in the final prediction models are similar using the two approaches.
2. Notably, applying a smoothing procedure effectively enhances prediction accuracy for distance-based approaches, evidenced by comparing the rows "LC-distance" versus "LC-distance-smooth" or "Multi-LC-distance, $K = 100$" versus "Multi-LC-distance-smooth, $K = 100$" in both female and male populations.
3. The last three rows in both female and male halves of Table 4 contain results of multiple-population distance-based approach implemented with a smoothing procedure and show the impact of a varying K, the maximum allowed number of "neighbors". Notably, the mean of the resulting 24 test SSEs exhibits a decreasing trend as K increases. In other words, enhanced predictive performance can be attained by expanding the candidate pool through an increase in K. However, it is important to acknowledge that the implementation of the multi-population distance-based method becomes computationally demanding as the value of K increases. This is precisely the motivation of proposing the ensemble method.
4. For male populations, the multi-population approaches lead to noticeable improvements in prediction accuracy compared to the single-population approaches. How-

ever, the benefit of borrowing information from the multiple populations is marginal for female populations.

Table 4: Summary statistics of test SSEs from the 24 populations comparing the prediction performance between distance-based models, Lee-Carter model, and age band based models.

	1st Quartile	Median	Mean	3rd Quartile
Female Population				
Lee-Carter	19.37	44.73	44.64	63.90
LC-ageband	17.44	44.49	44.00	67.36
LC-ageband-smooth	14.81	41.94	40.57	59.22
LC-distance	17.13	44.65	44.15	67.17
LC-distance-smooth	16.82	43.67	42.78	65.86
Multi-LC-distance, $K = 100$	16.50	41.91	45.78	72.24
Multi-LC-distance-smooth, $K = 100$	16.96	40.95	42.14	66.85
Multi-LC-distance-smooth, $K = 200$	17.39	40.86	40.58	62.25
Multi-LC-distance-smooth, $K = 400$	17.85	41.07	40.13	59.38
Male Population				
Lee-Carter	28.40	41.31	49.79	67.97
LC-ageband	16.78	37.67	39.14	60.66
LC-ageband-smooth	16.01	34.99	37.44	53.90
LC-distance	16.38	38.07	39.25	60.15
LC-distance-smooth	15.55	36.35	37.40	57.05
Multi-LC-distance, $K = 100$	15.43	36.09	37.14	57.02
Multi-LC-distance-smooth, $K = 100$	14.04	34.68	35.24	53.10
Multi-LC-distance-smooth, $K = 200$	14.50	33.91	34.79	51.98
Multi-LC-distance-smooth, $K = 400$	15.48	33.60	34.64	50.53

5.2 Impact of Hyperparameters in the Ensemble Method

In this section, we conduct numerical experiments to evaluate the performance of the proposed ensemble method and assess the influence of the hyperparameter triplet (n, m, N) on

its predictive performance. The results are summarized in Table 5 and Figure 6. We have the following observations.

- Excessively large values of the subsampling size n negatively impact predictive accuracy. This can be observed in Table 5 that there is a moderate increase in mean and median as n increases and m and N are fixed. The upper left panel of Figure 6 depicts an increasing trend of the average SSEs with respect to increasing values of n when the other two parameters are held constant at $m = 100$ and $N = 100$. Larger values of n can increase the chance of subsample overlapping and homogeneity among predictions from distinct subsamples, and eventually hurt the efficacy of the ensemble approach and cause the loss in prediction accuracy.
- The upper right panel of Figure 6 reveals that the optimal predictive accuracy is attained if the size of the selected age set m increases to the size of the subsample n. When $m = n$, the optimal choice for m and n lies between 100 and 200 as shown in the lower left panel.
- The curve in the lower right panel of Figure 6 suggests that changes in the subsampling instances N hardly affect predictive performance. This is consistent with the observation of no obvious changes and patterns in the last five rows of both panels in Table 5.

5.3 Performance Comparison Across Different Approaches

In this section, we compare the performance of the distance-based approach, the ensemble approach, and the ACF model based method. In Table 6, we report the results of the best performed distance-based approach "Multi-LC-distance-smooth, K = 400" from Table 4, the best performed ensemble approach with hyperparameters $(n, m, N) = (125, 100, 100)$ from Table 5, and add results of ACF model based approach. For ACF model based method, population selection is based on a geographical division given in Table 1 and age set selection is smoothed with $h = 4$.

The results reveal similar predicting accuracy between the ensemble method and the multi-population distance-based method, suggesting that the ensemble approach successfully maintains predictive power while being computationally more efficient. Moreover, the ACF model-based method (i.e., "ageband-ACFGeoInfo") shows a notable improvement in predicting accuracy over the other two approaches when predicting mortality for female populations, but not much for male populations.

Table 5: Summary statistics of test SSEs of the 24 populations comparing the prediction performance of ensemble approach with different values of hyperparameters, where N is the number of subsampling, n is the size of subsampled subset, and m is the "neighborhood" size.

Multi-LC-ensemble (n, m, N)	1st Quartile	Median	Mean	3rd Quartile
Female Population				
Multi-LC-ensemble $(125, 100, 100)$	18.51	39.33	40.98	62.01
Multi-LC-ensemble $(150, 100, 100)$	18.67	40.71	41.04	61.40
Multi-LC-ensemble $(175, 100, 100)$	18.80	42.71	41.22	61.08
Multi-LC-ensemble $(200, 100, 100)$	18.88	43.88	41.40	60.80
Multi-LC-ensemble $(200, 150, 100)$	18.91	43.99	41.32	60.32
Multi-LC-ensemble $(200, 100, 50)$	18.87	43.91	41.42	60.80
Multi-LC-ensemble $(200, 100, 200)$	18.88	43.90	41.40	60.78
Multi-LC-ensemble $(200, 100, 300)$	18.89	43.91	41.40	60.79
Multi-LC-ensemble $(200, 100, 400)$	18.88	43.90	41.41	60.78
Multi-LC-ensemble $(200, 100, 500)$	18.88	43.90	41.41	60.79
Male Population				
Multi-LC-ensemble $(125, 100, 100)$	15.09	33.86	34.69	52.04
Multi-LC-ensemble $(150, 100, 100)$	15.41	33.80	34.66	51.76
Multi-LC-ensemble $(175, 100, 100)$	15.79	33.62	34.68	51.77
Multi-LC-ensemble $(200, 100, 100)$	17.85	32.68	35.81	51.60
Multi-LC-ensemble $(200, 150, 100)$	16.17	33.24	34.65	51.19
Multi-LC-ensemble $(200, 100, 50)$	16.03	33.30	34.75	51.64
Multi-LC-ensemble $(200, 100, 200)$	15.99	33.34	34.73	51.72
Multi-LC-ensemble $(200, 100, 300)$	15.98	33.34	34.72	51.66
Multi-LC-ensemble $(200, 100, 400)$	15.98	33.32	34.72	51.69
Multi-LC-ensemble $(200, 100, 500)$	15.99	33.29	34.72	51.72

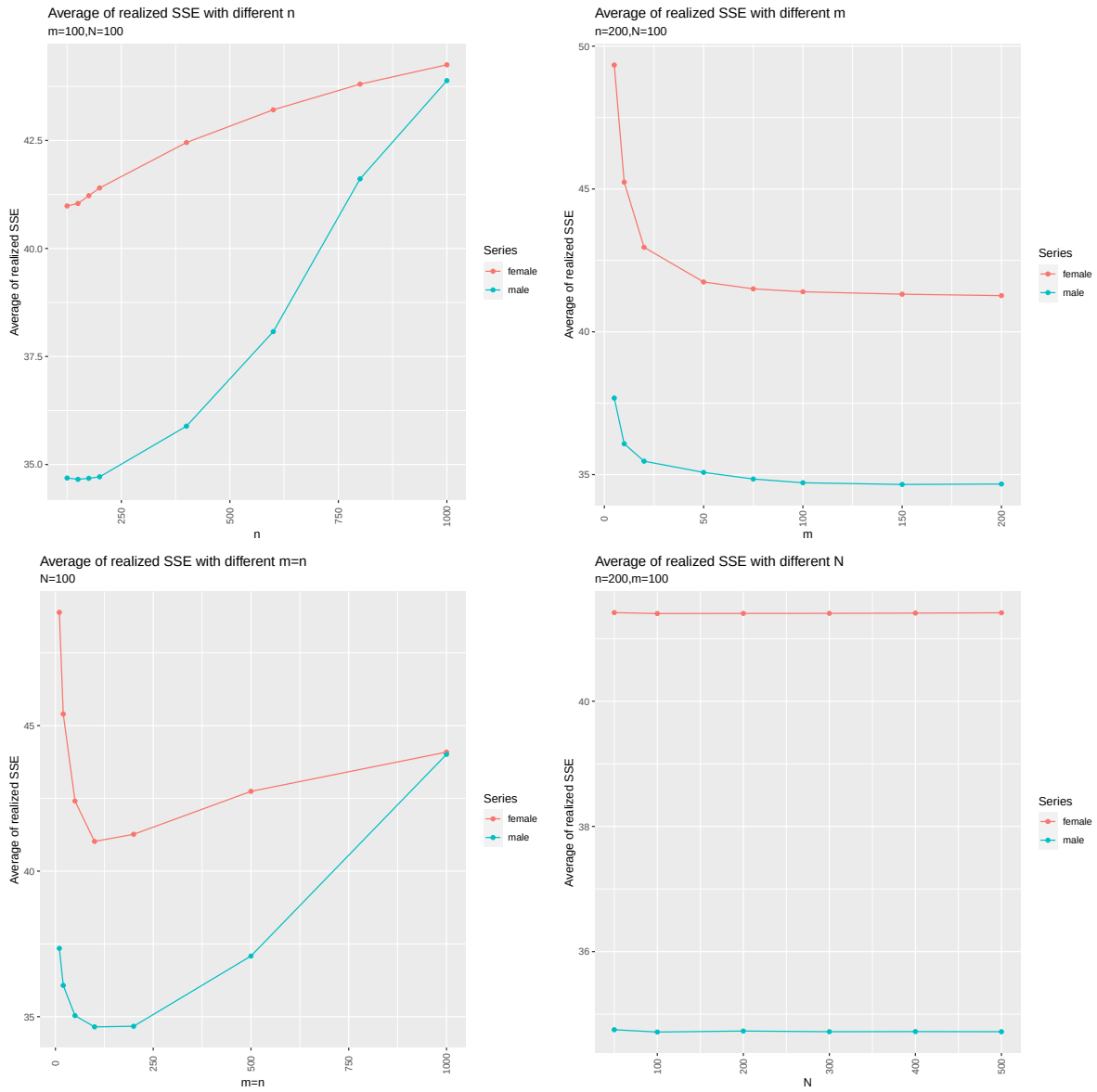


Figure 6: Average of test SSEs for the ensemble method with different values of hyperparameters. Top left: varying n and fixed m and N. Top right:varyingf m and fixed values of n and N. Bottom left: varying $m = n$ and fixed N. Bottom right: varing N and fixed m and n.

Table 6: Summary statistics of test SSEs of the 24 populations comparing the prediction performance of different proposed strategies.

	1st Quartile	Median	Mean	3rd Quartile
Female Population				
Multi-LC-distance-smooth, $K = 400$	17.85	41.07	40.13	59.38
Multi-LC-ensemble $(125, 100, 100)$	18.51	39.33	40.98	62.01
ageband-ACFGeoInfo	15.40	38.87	37.23	54.29
Male Population				
Multi-LC-distance-smooth, $K = 400$	15.48	33.60	34.64	50.53
Multi-LC-ensemble $(125, 100, 100)$	15.09	33.86	34.69	52.04
ageband-ACFGeoInfo	19.23	33.11	36.49	49.71

6 Concluding Remarks

Observations on similarities in mortality development trajectories of neighboring ages are widely made in mortality data and offer opportunities for enhancing mortality prediction accuracy. Conversely, dissimilarities across distinct age ranges, such as adult versus senior ages, are also noticeable. It is undesirable that many conventional approaches model all ages jointly by a generic model. In this paper, we introduce a framework which allows to train a customized model for each specific target age using information from a single population or multiple populations. This offers a new perspective on mortality forecasting and underscores the importance of incorporating age-specific information into extrapolative models to achieve improved forecasting performance. Moreover, the proposed framework is versatile to embed a wide range of existing mortality models. While the empirical studies in this paper focus on the utilization of the Lee-Carter model and the ACF model, the framework can accommodate various other mortality models.

References

Christoph Bergmeir and José M Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012.

Christoph Bergmeir, Mauro Costantini, and José M Benítez. On the usefulness of cross-

validation for directional forecast evaluation. *Computational Statistics & Data Analysis*, 76:132–143, 2014.

Heather Booth, John Maindonald, and Len Smith. Applying Lee-Carter under conditions of variable mortality decline. *Population Studies*, 56(3):325–336, 2002.

Andrew JG Cairns, David Blake, and Kevin Dowd. A two-factor model for stochastic mortality with parameter uncertainty: theory and calibration. *Journal of Risk and Insurance*, 73(4):687–718, 2006.

Andrew JG Cairns, David Blake, Kevin Dowd, Guy D Coughlan, David Epstein, Alen Ong, and Igor Balevich. A quantitative comparison of stochastic mortality models using data from england and wales and the united states. *North American Actuarial Journal*, 13(1): 1–35, 2009.

Malika Charrad, Nadia Ghazzali, Véronique Boiteau, and Azam Niknafs. Nbclust: an R package for determining the relevant number of clusters in a data set. *Journal of Statistical Software*, 61(1):1–36, 2014.

Piet De Jong and Leonie Tickle. Extending Lee-Carter mortality forecasting. *Mathematical Population Studies*, 13(1):1–18, 2006.

Liqun Diao, Yechao Meng, and Chengguo Weng. A DSA algorithm for mortality forecasting. *North American Actuarial Journal*, 25(3):438–458, 2021.

Liqun Diao, Yechao Meng, Chengguo Weng, and Tony Wirjanto. Enhancing mortality forecasting through bivariate model–based ensemble. *North American Actuarial Journal*, pages 1–20, 2023.

Francis X Diebold and Roberto S Mariano. Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13(3):134–144, 1995.

Evelyn Fix and Joseph Lawson Hodges. Discriminatory analysis. Nonparametric discrimination: Consistency properties. *International Statistical Review/Revue Internationale de Statistique*, 57(3):238–247, 1989.

Chris Fraley and Adrian E Raftery. Model-based clustering, discriminant analysis, and density estimation. *Journal of the American Statistical Association*, 97(458):611–631, 2002.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Rob J Hyndman, Heather Booth, and Farah Yasmeen. Coherent mortality forecasting: the product-ratio method with functional time series models. *Demography*, 50(1):261–283, 2013.

Salvatory R Kessy, Michael Sherris, Andrés M Villegas, and Jonathan Ziveyi. Mortality forecasting using stacked regression ensembles. *Scandinavian Actuarial Journal*, pages 1–36, 2021.

Torsten Kleinow. A common age effect model for the mortality of multiple populations. *Insurance: Mathematics and Economics*, 63:147–152, 2015.

Ronald Lee and Timothy Miller. Evaluating the performance of the Lee-Carter method for forecasting mortality. *Demography*, 38(4):537–549, 2001.

Ronald D Lee and Lawrence R Carter. Modeling and forecasting us mortality. *Journal of the American Statistical Association*, 87(419):659–671, 1992.

Jackie Li. A Poisson common factor model for projecting mortality and life expectancy jointly for females and males. *Population Studies*, 67(1):111–126, 2013.

Johnny Siu-Hang Li and Mary R Hardy. Measuring basis risk in longevity hedges. *North American Actuarial Journal*, 15(2):177–200, 2011.

Nan Li and Ronald Lee. Coherent mortality forecasts for a group of populations: An extension of the Lee-Carter method. *Demography*, 42(3):575–594, 2005.

Ronald Richman and Mario V Wuthrich. A neural network extension of the lee-carter model to multiple populations. *Available at SSRN 3270877*, 2018.

Maria Russolillo, Giuseppe Giordano, and Steven Haberman. Extending the Lee-Carter model: a three-way decomposition. *Scandinavian Actuarial Journal*, 2011(2):96–117, 2011.

Luca Scrucca, Michael Fop, T. Brendan Murphy, and Adrian E. Raftery. mclust 5: clustering, classification and density estimation using Gaussian finite mixture models. *The R Journal*, 8(1):289–317, 2016. URL `https://doi.org/10.32614/RJ-2016-021`.

Han Lin Shang and Steven Haberman. Retiree mortality forecasting: A partial age-range or a full age-range model? *Risks*, 8(3):69, 2020.

Han Lin Shang and Rob J Hyndman. Grouped functional time series forecasting: an application to age-specific mortality rates. *Journal of Computational and Graphical Statistics*, 26(2):330–343, 2017.

Dilan SriDaran, Michael Sherris, Andrés M Villegas, and Jonathan Ziveyi. A group regularisation approach for constructing generalised age-period-cohort mortality projection models. *ASTIN Bulletin*, 52(1):247–289, 2022.

Cary Chi-Liang Tsai and Echo Sihan Cheng. Incorporating statistical clustering methods into mortality models to improve forecasting performances. *Insurance: Mathematics and Economics*, 99:42–62, 2021.

John W Tukey et al. *Exploratory Data Analysis*, volume 2. Reading, MA, 1977.

Chou-Wen Wang, Jinggong Zhang, and Wenjun Zhu. Neighbouring prediction for mortality. *ASTIN Bulletin*, 51(3):689–718, 2021.

Joe H Ward Jr. Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301):236–244, 1963.

John R Wilmoth. Computational methods for fitting and extrapolating the Lee-Carter model of mortality change. Technical report, Technical report, Department of Demography, University of California, Berkeley, 1993.

Rui Zhou, Yujiao Wang, Kai Kaufhold, Johnny Siu-Hang Li, and Ken Seng Tan. Modeling period effects in multi-population mortality models: Applications to Solvency II. *North American Actuarial Journal*, 18(1):150–167, 2014.